AUTOSAR ADAPTIVE PLATFORM SPECIFICATION

High-Level Design Specification (HLD) & Interface Control Document

Next-Generation Electric Powertrain Domain Controller (ePDC)

SYNTHETIC ACADEMIC SPECIFICATION ? FOR BENCHMARK EVALUATION ONLY

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Academic & Synthetic Integrity Notice

This document is a synthetic, illustrative High-Level Design (HLD) engineering document created strictly for academic research, document intelligence benchmarking, and AI evaluation of the AUTOSAR AI platform. It is NOT an authentic OEM or commercial automotive document and does NOT claim official compliance with any commercial vehicle platform. All requirements, component architectures, port mappings, and performance budgets are simulated engineering assumptions.

Document Management & Governance

This High-Level Design specification governs the architectural decomposition, software component interfaces, scheduling paradigms, functional safety mechanisms, and execution profiles for the synthetic AUTOSAR Adaptive Platform Electric Powertrain Domain Controller (ePDC). All updates must follow strict document change governance.

Revision History

Rev	Date	Primary Author	Description of Changes	Approval
0.1	2026-01-15	Arch Team		Internal
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Engineering Authority Sign-Off Table

Role	Name	Department / Affiliation	Signature / Status	Sign-Off Date
Lead Systems Architect	Dr. K. S. Ramanathan		APPROVED	2026-10-18
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Software Release Lead	Abineswari P.		FINAL RELEASE	2026-10-19

Target Hardware Execution Environment

The reference target hardware platform comprises an automotive-grade multi-core System-on-Chip (SoC) featuring four ARM Cortex-A78AE application cores operating at 2.0 GHz running POSIX-compliant Adaptive Autosar OS (QNX Neutrino / Real-Time Linux), paired with dual ARM Cortex-R52 real-time lockstep safety processor cores operating at 800 MHz dedicated to the ASIL-D Emergency Cutoff and Watchdog supervisor.

Table of Contents (Chapters 1 to 23)

Ch.	Chapter Title	Primary Architectural Focus	Page Range
1			Pages 4?4
2			Pages 5?8
3			Pages 9?12
4	Functional Requirements Catalog		Pages 13?18
5			Pages 19?22
6			Pages 23?26
7			Pages 27?32
8			Pages 33?38
9			Pages 39?43
10			Pages 44?48
11			Pages 49?53
12			Pages 54?60
13			Pages 61?65
14			Pages 66?69
15			Pages 70?74
16			Pages 75?78
17			Pages 79?83
18			Pages 84?87
19			Pages 88?90
20			Pages 91?93
21			Pages 94?96
22			Pages 97?98
23			Pages 99?101

1.1 Document Scope and Purpose

This document defines the High-Level Design (HLD) specification for the synthetic Electric Powertrain Domain Controller (ePDC), an advanced Electronic Control Unit (ECU) implemented on the AUTOSAR Adaptive Platform. The primary objective of this architecture is to provide a deterministic, high-assurance software environment capable of coordinating electric traction motor torque, high-voltage battery state management, multi-loop thermal regulation, and vehicle motion stability while satisfying stringent ISO 26262 ASIL-D functional safety constraints.

1.2 Target Audience & Engineering Assumptions

This specification serves system architects, safety engineers, software developers, and verification teams. The software components specified herein execute within an Adaptive AUTOSAR runtime environment compliant with POSIX PSE52 profiles. Real-time safety monitors execute within a hardware-isolated lockstep companion core.

Explicit Architectural Boundary: Synthetic Nature

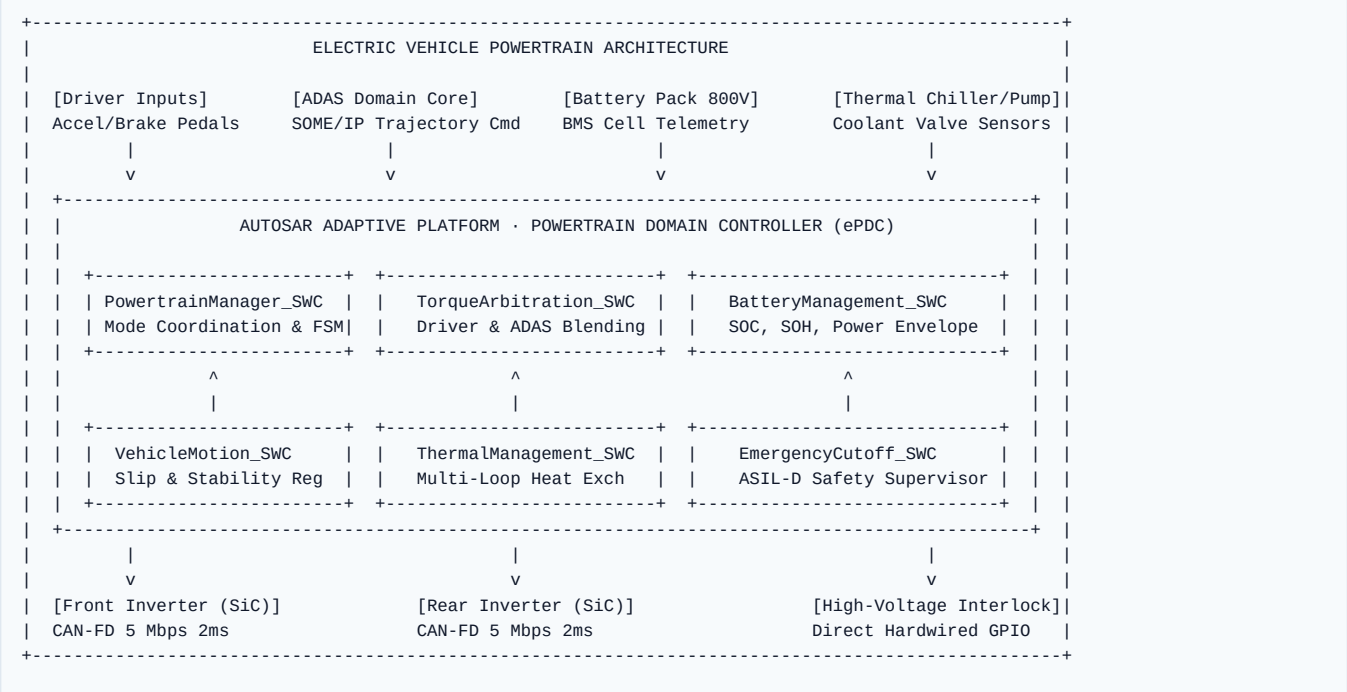
This entire specification is an engineering synthesis created for benchmarking automated RAG systems. It is not an actual production design. Operating voltages, communication cycle times, and torque blending equations represent typical automotive figures but are non-binding outside this academic evaluation framework.

Assumption ID	Architectural Impact	Safety Relevance
ASM-GEN-001		ASIL-D
ASM-GEN-002		ASIL-B
ASM-GEN-003		QM
ASM-GEN-004		ASIL-D

2.1 Electric Powertrain Domain Controller Overview

The Electric Powertrain Domain Controller (ePDC) functions as the central computation hub for electric propulsion in modern high-performance battery electric vehicles (BEVs). It directly arbitrates driver acceleration and deceleration intents, interacts with advanced driver assistance systems (ADAS) via service-oriented SOME/IP interfaces, manages energy flow to and from the high-voltage battery, and coordinates dynamic cooling loops across motors, inverters, and battery cells.

2.2 High-Level Physical Architecture Topology



The controller hosts six core Software Components (SWCs), two auxiliary controllers (InverterController_SWC, RegenerativeBraking_SWC), and an isolated ASIL-D safety monitor executing on a dedicated lockstep micro-core. The physical interfaces comprise automotive Gigabit Ethernet (1000BASE-T1) connecting to the vehicle backbone, dual CAN-FD channels connecting to motor inverters, and hardwired discrete digital pins for the high-voltage interlock line.

2.2 Operational States and System Modes

The ePDC software enforces a deterministic finite state machine governing powertrain lifecycle states. Under no circumstances may motor torque be delivered when the system is not in the RUN_ACTIVE state. Transitions between states require explicit multi-condition validation across battery pre-charge status, inverter ready flags, and safety interlock continuity.

State Name	Entry Conditions	Allowable Actions	Exit Triggers	Target Safe State
PWR_OFF			Clamp 15 wake-up pulse	PWR_OFF
PWR_STANDBY			Pre-charge complete or fault	PWR_OFF
PWR_PRECHARGE			Delta V < 10V within 500ms	PWR_FAULT
PWR_READY			Driver selects D/R gear	PWR_STANDBY
RUN_ACTIVE			Gear to P/N or fault detect	RUN_DEGRADED
RUN_DEGRADED			Critical fault or ignition off	PWR_SAFE_STOP
PWR_SAFE_STOP			Vehicle speed = 0 km/h	PWR_OFF
PWR_FAULT			Manual service reset only	PWR_OFF

Safety Requirement SR-PWR-010 (State Invariant)

The transition from RUN_ACTIVE to PWR_SAFE_STOP must initiate inverter zero-torque control within 10 ms of fault assertion. If inverters fail to respond within 20 ms, the EmergencyCutoff_SWC shall assert the hardware interlock to physically decouple the high-voltage contactors.

2.3 High-Voltage Energy Architecture

The vehicle electrical architecture utilizes an 800V nominal lithium-ion battery pack with a total energy capacity of 96 kWh, constructed from 192 cells in series (2P192S configuration). The battery delivers continuous electrical power of up to 350 kW to front and rear dual-inverter traction drives, and absorbs up to 250 kW during high-deceleration regenerative braking events.

Power Distribution Specifications

Subsystem Parameter	Nominal Value	Minimum Operating Limit		Overvoltage Cutoff
Pack Voltage	800 V DC	620 V DC (0% SOC)	890 V DC (100% SOC)	920 V DC (Instant Cutoff)
Peak Discharge Current	500 A (10 seconds)	0 A	450 A (Continuous)	600 A (Pyrofuse Trigger)
Peak Regenerative Current	320 A (10 seconds)	0 A	250 A (Continuous)	350 A (Software Clamp)
Front Motor Power	160 kW continuous	0 kW	200 kW peak (5 sec)	210 kW hard limit
Rear Motor Power	220 kW continuous	0 kW	280 kW peak (5 sec)	300 kW hard limit
	270 kW peak	10 kW	150 kW average (10-80%)	300 kW peak pulse

To prevent catastrophic thermal runaway and cell damage, the BatteryManagement_SWC computes continuous real-time discharge and charge power limits ($P_{discharge_max}$, P_{charge_max}) at a 10 ms cyclic rate. These limits are transmitted via SOME/IP service interfaces to the TorqueArbitration_SWC, which clips the requested motor torque accordingly.

2.4 Dual Electric Drive Subsystem (Front & Rear Motors)

The powertrain implements an all-wheel-drive (AWD) torque-vectoring architecture featuring an interior permanent magnet synchronous motor (IPMSM) on the rear axle and an induction asynchronous motor (ASM) on the front axle. This combination provides high efficiency during steady cruising (by de-energizing the front induction motor with zero drag losses) and immediate high torque response during dynamic acceleration.

Drive Attribute	Front Axle Motor (Induction)		Combined AWD System
Motor Topology	3-phase AC Induction Motor (ASM)		Hybrid Dual-Technology AWD
Peak Torque	280 Nm @ 0?4500 RPM	450 Nm @ 0?5500 RPM	730 Nm Wheel-Equivalent
Continuous Torque	180 Nm	320 Nm	500 Nm Continuous
Max Rotor Speed	16,000 RPM	18,000 RPM	Electronically Synchronized
Inverter Technology	Silicon Carbide (SiC) MOSFET	Silicon Carbide (SiC) MOSFET	Dual Inverter Control
Cooling Medium	Dielectric Oil Direct Spray	Dielectric Oil + Water Glycol Jacket	Integrated Multi-Loop Chiller
Inverter Interface	CAN-FD Bus 1 (2 ms cycle)	CAN-FD Bus 2 (2 ms cycle)	Dual Redundant Channels

Torque distribution between the front and rear axles is calculated by VehicleMotion_SWC based on dynamic axle load transfer, steering angle, yaw rate error, and tire slip ratios. The resulting torque targets (T_{cmd_front}, T_{cmd_rear}) are forwarded to TorqueArbitration_SWC for safety envelope compliance checks.

3.1 Vehicle System Context Diagram

The Electric Powertrain Domain Controller does not operate in isolation; it interfaces with six major vehicle electronic control units across three heterogeneous bus networks: Deterministic Automotive Ethernet (1000BASE-T1), High-Speed CAN-FD (5 Mbps), and hardwired safety-critical discrete circuits.

Interfacing ECU / Actor	Bus Type & Protocol	Cycle Rate	Payload Description	ASIL Level
Body Control Module (BCM)	Ethernet / SOME/IP	100 ms		ASIL-B
ADAS Central Computer	Ethernet / SOME/IP	10 ms		ASIL-D
	CAN-FD / SecOC	10 ms		ASIL-D
Steering Control Unit (EPS)	CAN-FD / SecOC	10 ms		ASIL-B
Front Motor Inverter	Dedicated CAN-FD 1	2 ms		ASIL-D
Rear Motor Inverter	Dedicated CAN-FD 2	2 ms		ASIL-D
	Ethernet / SOME/IP	50 ms		ASIL-C
Crash Pyrofuse Actuator	Hardwired Discrete	Continuous		ASIL-D

3.2 Hardware-Software Interface (HSI)

The ePDC software interacts with hardware peripherals via standard AUTOSAR Adaptive Microcontroller Abstraction and POSIX IO drivers. The physical hardware provides hardware security modules (HSM) for cryptographic acceleration and hardware timers for sub-microsecond timestamping.

Physical Resource	Hardware Specification	Mapped AUTOSAR Driver	Access Restrictions	Driver Priority
Ethernet MAC 0		EthIf / ComM Driver	Ring 0 Adaptive Kernel	High
CAN-FD Controller 1	Bosch M_CAN IP / 5 Mbps	CanIf / CAN Driver	POSIX Character Device	High
CAN-FD Controller 2	Bosch M_CAN IP / 5 Mbps	CanIf / CAN Driver	POSIX Character Device	High
Hardware Security Module	Evita Full / Secure Core	CryptoStack / ARA::CRY	Protected Secure Enclave	Critical
Flash Memory (eMMC/UFS)	128 GB UFS 3.1 / SLC Pool	Persistency / ARA::PER	Encrypted Partition Only	Normal
Hardware Watchdog		Watchdog Driver	Lockstep Core Direct Line	Real-Time

3.3 Stakeholder Registry and Architectural Influence

Stakeholder needs directly shape the structural partitioning, verification criteria, and quality metrics of the ePDC software architecture. The table below formalizes primary stakeholders, their operational concerns, and architectural design mitigations.

Stakeholder Group	Primary Concerns & Objectives	Architectural Response	Verification Standard
Vehicle End Users			ISO 2631 Ride Comfort
Functional Safety Auditors			ISO 26262 Part 4/6
Homologation Authorities			UNECE R155 / R156
Fleet & Service Technicians			ISO 14229 / ISO 13400
Manufacturing / Plant			Automotive SPICE Level 3

3.4 Environmental Operating Envelope

The ePDC enclosure is mounted in the vehicle under-hood powertrain bay and must operate continuously within harsh ambient thermal, mechanical, and electromagnetic environments. Software components must dynamically adjust execution profiles according to ambient environmental feedback.

Environmental Parameter	Nominal Range	Extreme Lower Limit	Extreme Upper Limit	Software Derating Action
Ambient Enclosure Temp	-40°C to +85°C	-45°C (Cold Soak)	+105°C (Extreme Soak)	
Coolant Inflow Temp	+25°C to +45°C	-30°C	+70°C	
Supply Voltage (12V Bus)	12.0 V to 14.4 V	6.5 V (Cranking)	18.0 V (Overvoltage)	
Mechanical Vibration	ISO 16750-3 Profile IV	0 Hz	2000 Hz random vibration	
Relative Humidity	5% to 95% non-cond	0%	100% condensing	

4.1 Functional Requirements Group 1

This section formalizes functional requirements REQ-PWR-001 through REQ-PWR-007. Each requirement specifies exact operational constraints, functional safety classification, and responsible software component implementing the requirement.

Req ID	Requirement Title	ASIL	Responsible SWC
REQ-PWR-001		ASIL-D	TorqueArbitration_SWC
REQ-PWR-002		ASIL-D	EmergencyCutoff_SWC
REQ-PWR-003		ASIL-B	TorqueArbitration_SWC
REQ-PWR-004		QM	PowertrainManager_SWC
REQ-PWR-005		ASIL-D	RegenerativeBraking_SWC
REQ-PWR-006		ASIL-D	BatteryManagement_SWC
REQ-PWR-007		ASIL-C	ThermalManagement_SWC

Requirements Verification Note for Group 1

All requirements in Group 1 must be verified using Hardware-in-the-Loop (HIL) fault injection test suites with automated pass/fail assertion triggers recorded in the compliance database.

4.2 Functional Requirements Group 2

This section formalizes functional requirements REQ-PWR-008 through REQ-PWR-014. Each requirement specifies exact operational constraints, functional safety classification, and responsible software component implementing the requirement.

Req ID	Requirement Title	ASIL	Responsible SWC
REQ-PWR-008		ASIL-D	BatteryManagement_SWC
REQ-PWR-009	Contactor Weld Detection	ASIL-D	BatteryManagement_SWC
REQ-PWR-010		ASIL-D	EmergencyCutoff_SWC
REQ-PWR-011		ASIL-D	EmergencyCutoff_SWC
REQ-PWR-012		QM	ThermalManagement_SWC
REQ-PWR-013		ASIL-C	InverterController_SWC
REQ-PWR-014	Motor Stall Protection	ASIL-B	TorqueArbitration_SWC

Requirements Verification Note for Group 2

All requirements in Group 2 must be verified using Hardware-in-the-Loop (HIL) fault injection test suites with automated pass/fail assertion triggers recorded in the compliance database.

4.3 Functional Requirements Group 3

This section formalizes functional requirements REQ-PWR-015 through REQ-PWR-021. Each requirement specifies exact operational constraints, functional safety classification, and responsible software component implementing the requirement.

Req ID	Requirement Title	ASIL	Responsible SWC
REQ-PWR-015		ASIL-B	VehicleMotion_SWC
REQ-PWR-016		ASIL-D	VehicleMotion_SWC
REQ-PWR-017	Reverse Gear Speed Limiter	ASIL-B	PowertrainManager_SWC
REQ-PWR-018		ASIL-B	TorqueArbitration_SWC
REQ-PWR-019		ASIL-C	PowertrainManager_SWC
REQ-PWR-020		QM	PowertrainManager_SWC
REQ-PWR-021		ASIL-B	BatteryManagement_SWC

Requirements Verification Note for Group 3

All requirements in Group 3 must be verified using Hardware-in-the-Loop (HIL) fault injection test suites with automated pass/fail assertion triggers recorded in the compliance database.

4.4 Functional Requirements Group 4

This section formalizes functional requirements REQ-PWR-022 through REQ-PWR-028. Each requirement specifies exact operational constraints, functional safety classification, and responsible software component implementing the requirement.

Req ID	Requirement Title	ASIL	Responsible SWC
REQ-PWR-022		ASIL-D	BatteryManagement_SWC
REQ-PWR-023		ASIL-B	ThermalManagement_SWC
REQ-PWR-024		ASIL-C	RegenerativeBraking_SWC
REQ-PWR-025	Creep Torque Control	QM	TorqueArbitration_SWC
REQ-PWR-026		ASIL-B	BatteryManagement_SWC
REQ-PWR-027		ASIL-D	EmergencyCutoff_SWC
REQ-PWR-028		ASIL-B	TorqueArbitration_SWC

Requirements Verification Note for Group 4

All requirements in Group 4 must be verified using Hardware-in-the-Loop (HIL) fault injection test suites with automated pass/fail assertion triggers recorded in the compliance database.

4.5 Functional Requirements Group 5

This section formalizes functional requirements REQ-PWR-029 through REQ-PWR-034. Each requirement specifies exact operational constraints, functional safety classification, and responsible software component implementing the requirement.

Req ID	Requirement Title	ASIL	Responsible SWC
REQ-PWR-029		ASIL-D	InverterController_SWC
REQ-PWR-030		ASIL-D	TorqueArbitration_SWC
REQ-PWR-031		ASIL-B	UpdateConfigManager
REQ-PWR-032		QM	DiagnosticManager
REQ-PWR-033		QM	ExecutionManager
REQ-PWR-034		QM	InverterController_SWC

Requirements Verification Note for Group 5

All requirements in Group 5 must be verified using Hardware-in-the-Loop (HIL) fault injection test suites with automated pass/fail assertion triggers recorded in the compliance database.

4.6 Functional Requirements Group 6

This section formalizes functional requirements REQ-PWR-035 through REQ-PWR-040. Each requirement specifies exact operational constraints, functional safety classification, and responsible software component implementing the requirement.

Req ID	Requirement Title	ASIL	Responsible SWC
REQ-PWR-035		ASIL-C	PowertrainManager_SWC
REQ-PWR-036		ASIL-B	ThermalManagement_SWC
REQ-PWR-037		QM	BatteryManagement_SWC
REQ-PWR-038		ASIL-B	TorqueArbitration_SWC
REQ-PWR-039		ASIL-C	PowertrainManager_SWC
REQ-PWR-040		ASIL-D	EmergencyCutoff_SWC

Requirements Verification Note for Group 6

All requirements in Group 6 must be verified using Hardware-in-the-Loop (HIL) fault injection test suites with automated pass/fail assertion triggers recorded in the compliance database.

5.1 Deterministic Timing & Real-Time Deadlines

The Electric Powertrain Domain Controller executes hard and soft real-time software tasks. Hard real-time tasks (such as torque arbitration, battery safety cutoffs, and inverter communication) must never miss a single scheduling deadline. The table below formalizes worst-case execution time (WCET) limits, scheduling jitter, and end-to-end latency targets.

Timing Metric ID	Nominal Period	Max Allowable Jitter	Max Allowable WCET	End-to-End Budget
TIM-PWR-001	10 ms cyclic	± 0.25 ms	1.8 ms	10.0 ms maximum
TIM-PWR-002	Interrupt driven	< 0.05 ms	0.4 ms	10.0 ms hard limit
TIM-PWR-003	10 ms cyclic	± 0.50 ms	2.2 ms	15.0 ms maximum
TIM-PWR-004	1 ms cyclic	± 0.02 ms	0.15 ms	2.0 ms maximum
TIM-PWR-005	100 ms cyclic	± 5.0 ms	8.5 ms	200.0 ms maximum
TIM-PWR-006	Asynchronous	N/A	15.0 ms	50.0 ms maximum

5.2 Safety Integrity Levels & Reliability Metrics

In compliance with ISO 26262:2018 Road Vehicles ? Functional Safety, the overall powertrain safety goal 'Prevention of Unintended Vehicle Acceleration or Deceleration' is assigned the highest safety integrity level: ASIL-D. Software components directly controlling torque arbitration and high-voltage power paths inherit ASIL-D development rigor.

Safety / Reliability Metric	Target Metric Value	Applicable Standard	Design Mitigation Mechanism
Random Hardware Failure Rate	< 10 FIT (10^-8 failures/hour)	ISO 26262 ASIL-D Target	
Single Point Fault Metric (SPFM)	>= 99.0%	ISO 26262 ASIL-D Target	
Latent Fault Metric (LFM)	>= 90.0%	ISO 26262 ASIL-D Target	
	> 150,000 operating hours	Automotive SPICE Level 3	
Flash Data Retention	15 years @ 85°C ambient	JEDEC JESD22-A117	
Safety Loop Diagnostic Coverage	>= 99.5%	ISO 26262 ASIL-D Target	

5.3 Computational Resource Allocation Budgets

To prevent resource starvation, CPU core saturation, and memory overflow, strict static resource quotas are assigned to each Software Component and platform service. Under peak operating conditions (including simultaneous CAN-FD bursts, SOME/IP service discovery, and rapid torque transient calculations), aggregate CPU utilization must remain below 70%.

	CPU Core Affinity	Max CPU Budget	RAM Budget (SRAM)	Flash ROM Budget	Stack Reserve
	Core 0 (Cortex-A78AE)	12.0%	8.0 MB	16.0 MB	256 KB
	Core 1 (Cortex-A78AE)	18.5%	12.0 MB	24.0 MB	512 KB
	Core 0 (Cortex-A78AE)	10.0%	6.0 MB	12.0 MB	256 KB
	Core 2 (Cortex-A78AE)	6.0%	4.0 MB	8.0 MB	128 KB
VehicleMotion_SWC	Core 1 (Cortex-A78AE)	14.0%	10.0 MB	18.0 MB	512 KB
		4.0%	2.0 MB	4.0 MB	128 KB
	Core 2 & Core 3	15.0%	32.0 MB	64.0 MB	1024 KB
	All Cores Available	69.5% Peak	74.0 MB / 256 MB	146 MB / 512 MB	Healthy Safety Margin

5.4 EMC Resilience & Cybersecurity Performance

High-voltage silicon carbide switching in nearby traction inverters introduces substantial high-frequency electromagnetic interference (EMI). The ePDC communication architecture incorporates hardware shielding and digital packet filtering to maintain zero bit-error rates during 200 A/microsecond switching events.

Standard / Threat Model	Testing Requirement	Acceptance Criteria	Software Countermeasure
CISPR 25 Class 5	Radiated and conducted emissions	No spurious interrupt generation	
ISO 11452-2 / ISO 11452-4			
ISO 7637-2 / ISO 16750-2	Transient electrical conduction	Zero reset on power dip to 6.5V	
ISO/SAE 21434 CAL-4			
Fuzzing Stress Testing	100M malformed SOME/IP packets	Zero kernel crash or buffer overflow	

6.1 AUTOSAR Adaptive Runtime for Applications (ARA)

The AUTOSAR Adaptive Platform is engineered for compute-intensive distributed automotive domains requiring dynamic configuration and high-bandwidth service-oriented communication. Unlike the static OSEK/AUTOSAR Classic environment, Adaptive AUTOSAR applications run as isolated POSIX processes utilizing C++14/C++17 standard libraries and interacting with the system through Adaptive Foundation APIs (ara::core, ara::com, ara::exec, ara::per, ara::diag, ara::log, and ara::crypto).

ARA Functional Cluster	Standard Namespace	Primary Service Provided	Implementation Paradigm
Communication Management	ara::com		Proxy-Skeleton abstraction pattern
Execution Management	ara::exec		
Persistency Management	ara::per		Atomic multi-block write with journal
Diagnostic Management	ara::diag		
Logging & Tracing	ara::log		
Cryptography Service	ara::crypto		
Update & Config Manager	ara::ucm		

6.2 Manifests, Software Modeling & Machine Deployment

Adaptive AUTOSAR strictly decouples application software design from physical deployment through standardized ARXML manifest files. The deployment workflow involves three distinct manifest tiers: Execution Manifests, Service Instance Manifests, and Machine Manifests.

Manifest Category	File Format & Extension	Validation Tooling
Execution Manifest	application_exec.arxml	AUTOSAR Manifest Validator
Service Instance Manifest	service_instance.arxml	Network Design Tool Suite
Machine Manifest	machine_manifest.arxml	Platform Integration Tool
Software Package Manifest	sw_package.arxml	UCM Packaging Utility

6.3 POSIX PSE52 Compliance & Kernel Separation

The underlying operating system provides POSIX PSE52 (Real-Time Controller) conformance with multi-threading, memory protection (MMU), real-time priority-preemptive scheduling (SCHED_FIFO), and process isolation. Software Components execute entirely within user-space memory partitions with no direct access to physical memory addresses.

POSIX Feature Set	Standard Reference	Kernel Implementation	Safety / Security Impact
Memory Management	POSIX mman.h / mprotect()		
Thread Scheduling	POSIX pthread / SCHED_FIFO		
Inter-Process Comms	POSIX shm_open() / mmap()		
Timers and Clocks	CLOCK_MONOTONIC_RAW		
Signal Handling	POSIX sigaction()		

6.4 Identity & Access Management (IAM) and Process Sandboxing

The Adaptive Platform enforces Identity and Access Management (IAM) to verify whether a given application process possesses authorization to offer or consume a specific service interface, open a communication socket, or access a persistency file. Permissions are evaluated statically at deployment time and verified dynamically at runtime.

Process Name	Assigned POSIX UID			
PowertrainManager	uid: 1001 (powertrain)			rw: /var/data/pwr_mgr/
TorqueArbitration	uid: 1002 (torque)			rw: /var/data/torque/
BatteryManagement	uid: 1003 (bms)	IBatteryStatus (Svc 0x1003)		rw: /var/data/bms/
ThermalManagement	uid: 1004 (thermal)			rw: /var/data/thermal/
VehicleMotion	uid: 1005 (motion)		IPowertrainCoordination	rw: /var/data/motion/
EmergencyCutoff	uid: 1000 (safety_root)			r: /proc/, rw: /dev/safety_io

7.1 PowertrainManager_SWC

Coordinates high-level vehicle drive modes, ignition clamp states, driver gear selections, and cruise control interfaces. It arbitrates the operational mode of the vehicle and coordinates auxiliary controllers during startup and shutdown.

Component Interfaces & Port Catalog

Port Name	Port Direction	Interface Type	Transport Binding	Update Rate
	Provided (PPort)	IPowertrainManagerStatus	SOME/IP Broadcast	10 ms cyclic
RPort_VehicleBusData	Required (RPort)	IVehicleBusSignals	SOME/IP Consume	10 ms cyclic
PPort_DiagnosticEvents	Provided (PPort)	IDiagnosticEventReport	ara::diag Direct	On Event
RPort_SafetyInterlocks	Required (RPort)	IEmergencySafetyLoop	Shared Memory / Direct	1 ms cyclic

Component Structural Invariants

Invariant ID	Invariant Mathematical Condition	Enforcement Action	Safety Class
INV-POW-001	Execution time < 2.0 ms per cycle		ASIL-B
INV-POW-002			ASIL-D
INV-POW-003		Trigger graceful recovery routine	ASIL-D

7.2 TorqueArbitration_SWC

Performs deterministic blending and arbitration between driver accelerator pedal requests, ADAS autonomous driving commands, traction control cutbacks, and stability interventions. Ensures total commanded torque does not exceed battery or motor limits.

Component Interfaces & Port Catalog

Port Name	Port Direction	Interface Type	Transport Binding	Update Rate
	Provided (PPort)	ITorqueArbitrationStatus	SOME/IP Broadcast	10 ms cyclic
RPort_VehicleBusData	Required (RPort)	IVehicleBusSignals	SOME/IP Consume	10 ms cyclic
PPort_DiagnosticEvents	Provided (PPort)	IDiagnosticEventReport	ara::diag Direct	On Event
RPort_SafetyInterlocks	Required (RPort)	IEmergencySafetyLoop	Shared Memory / Direct	1 ms cyclic

Component Structural Invariants

Invariant ID	Invariant Mathematical Condition	Enforcement Action	Safety Class
INV-TOR-001	Execution time < 2.0 ms per cycle		ASIL-B
INV-TOR-002			ASIL-D
INV-TOR-003		Trigger graceful recovery routine	ASIL-D

7.3 BatteryManagement_SWC

Calculates state of charge (SOC), state of health (SOH), maximum continuous discharge power, and maximum regenerative charge acceptance. Monitors individual cell voltages and temperatures, controlling pre-charge sequences and contactors.

Component Interfaces & Port Catalog

Port Name	Port Direction	Interface Type	Transport Binding	Update Rate
	Provided (PPort)	IBatteryManagementStatus	SOME/IP Broadcast	10 ms cyclic
RPort_VehicleBusData	Required (RPort)	IVehicleBusSignals	SOME/IP Consume	10 ms cyclic
PPort_DiagnosticEvents	Provided (PPort)	IDiagnosticEventReport	ara::diag Direct	On Event
RPort_SafetyInterlocks	Required (RPort)	IEmergencySafetyLoop	Shared Memory / Direct	1 ms cyclic

Component Structural Invariants

Invariant ID	Invariant Mathematical Condition	Enforcement Action	Safety Class
INV-BAT-001	Execution time < 2.0 ms per cycle		ASIL-B
INV-BAT-002			ASIL-D
INV-BAT-003		Trigger graceful recovery routine	ASIL-D

7.4 ThermalManagement_SWC

Controls electric coolant pumps, proportional flow valves, front radiator shutters, and air conditioning refrigerant chillers to regulate battery, motor, and inverter temperatures within optimal thermal efficiency bands.

Component Interfaces & Port Catalog

Port Name	Port Direction	Interface Type	Transport Binding	Update Rate
	Provided (PPort)	IThermalManagementStatus	SOME/IP Broadcast	10 ms cyclic
RPort_VehicleBusData	Required (RPort)	IVehicleBusSignals	SOME/IP Consume	10 ms cyclic
PPort_DiagnosticEvents	Provided (PPort)	IDiagnosticEventReport	ara::diag Direct	On Event
RPort_SafetyInterlocks	Required (RPort)	IEmergencySafetyLoop	Shared Memory / Direct	1 ms cyclic

Component Structural Invariants

Invariant ID	Invariant Mathematical Condition	Enforcement Action	Safety Class
INV-THE-001	Execution time < 2.0 ms per cycle		ASIL-B
INV-THE-002			ASIL-D
INV-THE-003		Trigger graceful recovery routine	ASIL-D

7.5 VehicleMotion_SWC

Evaluates steering angle, wheel speed sensors, lateral acceleration, and yaw rate to optimize torque split between front and rear axles. Mitigates understeer and oversteer through active torque vectoring.

Component Interfaces & Port Catalog

Port Name	Port Direction	Interface Type	Transport Binding	Update Rate
PPort_VehicleMotionStatus	Provided (PPort)	IVehicleMotionStatus	SOME/IP Broadcast	10 ms cyclic
RPort_VehicleBusData	Required (RPort)	IVehicleBusSignals	SOME/IP Consume	10 ms cyclic
PPort_DiagnosticEvents	Provided (PPort)	IDiagnosticEventReport	ara::diag Direct	On Event
RPort_SafetyInterlocks	Required (RPort)	IEmergencySafetyLoop	Shared Memory / Direct	1 ms cyclic

Component Structural Invariants

Invariant ID	Invariant Mathematical Condition	Enforcement Action	Safety Class
INV-VEH-001	Execution time < 2.0 ms per cycle		ASIL-B
INV-VEH-002			ASIL-D
INV-VEH-003		Trigger graceful recovery routine	ASIL-D

7.6 EmergencyCutoff_SWC

Executes isolated plausibility monitoring on a companion lockstep core. Monitors commanded versus actual torque, accelerator versus brake conflicts, and HVIL continuity. Triggers hardware cutoff if hazardous torque is detected.

Component Interfaces & Port Catalog

Port Name	Port Direction	Interface Type	Transport Binding	Update Rate
	Provided (PPort)	IEmergencyCutoffStatus	SOME/IP Broadcast	10 ms cyclic
RPort_VehicleBusData	Required (RPort)	IVehicleBusSignals	SOME/IP Consume	10 ms cyclic
PPort_DiagnosticEvents	Provided (PPort)	IDiagnosticEventReport	ara::diag Direct	On Event
RPort_SafetyInterlocks	Required (RPort)	IEmergencySafetyLoop	Shared Memory / Direct	1 ms cyclic

Component Structural Invariants

Invariant ID	Invariant Mathematical Condition	Enforcement Action	Safety Class
INV-EME-001	Execution time < 2.0 ms per cycle		ASIL-B
INV-EME-002			ASIL-D
INV-EME-003		Trigger graceful recovery routine	ASIL-D

8.1 PowertrainManager_SWC State Machine

This section documents the formal mathematical formulations, internal state machine definitions, and execution invariants governing PowertrainManager_SWC State Machine. All equations are executed cyclically by the component's internal runnable entities.

Detailed Algorithmic Specification

Torque request calculation combines driver pedal mapping, cruise control PID output, and speed limit governor commands: T_raw = (Pedal_Ratio * T_max_curve(RPM)) + T_cruise_delta. The result is clipped against the battery power envelope: T_clipped = min(T_raw, (P_battery_discharge_limit / Motor_Angular_Velocity)). When friction braking is active, regenerative torque is blended smoothly: T_final = T_clipped - T_regen_target.

Algorithmic Step	Execution Frequency	Expected Output Range
Step 1: Input Conditioning	10 ms	0.0% to 100.0%
Step 2: Base Torque Lookup	10 ms	-150 Nm to +730 Nm
Step 3: Power Limit Clamp	10 ms	-250 Nm to +730 Nm
Step 4: Stability Adjustment	10 ms	Front: 0-280Nm, Rear: 0-450Nm
Step 5: Safety Verification	2 ms	Boolean Invariant Check

Safety Invariant Verification

Any violation of Step 5 where commanded torque exceeds actual motor torque capability by more than 50 Nm for longer than 20 ms triggers immediate ASIL-D safe stop initiation.

8.2 TorqueArbitration_SWC Blending Math

This section documents the formal mathematical formulations, internal state machine definitions, and execution invariants governing TorqueArbitration_SWC Blending Math. All equations are executed cyclically by the component's internal runnable entities.

Detailed Algorithmic Specification

Torque request calculation combines driver pedal mapping, cruise control PID output, and speed limit governor commands: T_raw = (Pedal_Ratio * T_max_curve(RPM)) + T_cruise_delta. The result is clipped against the battery power envelope: T_clipped = min(T_raw, (P_battery_discharge_limit / Motor_Angular_Velocity)). When friction braking is active, regenerative torque is blended smoothly: T_final = T_clipped - T_regen_target.

Algorithmic Step	Execution Frequency	Expected Output Range
Step 1: Input Conditioning	10 ms	0.0% to 100.0%
Step 2: Base Torque Lookup	10 ms	-150 Nm to +730 Nm
Step 3: Power Limit Clamp	10 ms	-250 Nm to +730 Nm
Step 4: Stability Adjustment	10 ms	Front: 0-280Nm, Rear: 0-450Nm
Step 5: Safety Verification	2 ms	Boolean Invariant Check

Safety Invariant Verification

Any violation of Step 5 where commanded torque exceeds actual motor torque capability by more than 50 Nm for longer than 20 ms triggers immediate ASIL-D safe stop initiation.

8.3 BatteryManagement_SWC State Estimation

This section documents the formal mathematical formulations, internal state machine definitions, and execution invariants governing BatteryManagement_SWC State Estimation. All equations are executed cyclically by the component's internal runnable entities.

Detailed Algorithmic Specification

Torque request calculation combines driver pedal mapping, cruise control PID output, and speed limit governor commands: T_raw = (Pedal_Ratio * T_max_curve(RPM)) + T_cruise_delta. The result is clipped against the battery power envelope: T_clipped = min(T_raw, (P_battery_discharge_limit / Motor_Angular_Velocity)). When friction braking is active, regenerative torque is blended smoothly: T_final = T_clipped - T_regen_target.

Algorithmic Step	Execution Frequency	Expected Output Range
Step 1: Input Conditioning	10 ms	0.0% to 100.0%
Step 2: Base Torque Lookup	10 ms	-150 Nm to +730 Nm
Step 3: Power Limit Clamp	10 ms	-250 Nm to +730 Nm
Step 4: Stability Adjustment	10 ms	Front: 0-280Nm, Rear: 0-450Nm
Step 5: Safety Verification	2 ms	Boolean Invariant Check

Safety Invariant Verification

Any violation of Step 5 where commanded torque exceeds actual motor torque capability by more than 50 Nm for longer than 20 ms triggers immediate ASIL-D safe stop initiation.

8.4 ThermalManagement_SWC Control Loops

This section documents the formal mathematical formulations, internal state machine definitions, and execution invariants governing ThermalManagement_SWC Control Loops. All equations are executed cyclically by the component's internal runnable entities.

Detailed Algorithmic Specification

Torque request calculation combines driver pedal mapping, cruise control PID output, and speed limit governor commands: T_raw = (Pedal_Ratio * T_max_curve(RPM)) + T_cruise_delta. The result is clipped against the battery power envelope: T_clipped = min(T_raw, (P_battery_discharge_limit / Motor_Angular_Velocity)). When friction braking is active, regenerative torque is blended smoothly: T_final = T_clipped - T_regen_target.

Algorithmic Step	Execution Frequency	Expected Output Range
Step 1: Input Conditioning	10 ms	0.0% to 100.0%
Step 2: Base Torque Lookup	10 ms	-150 Nm to +730 Nm
Step 3: Power Limit Clamp	10 ms	-250 Nm to +730 Nm
Step 4: Stability Adjustment	10 ms	Front: 0-280Nm, Rear: 0-450Nm
Step 5: Safety Verification	2 ms	Boolean Invariant Check

Safety Invariant Verification

Any violation of Step 5 where commanded torque exceeds actual motor torque capability by more than 50 Nm for longer than 20 ms triggers immediate ASIL-D safe stop initiation.

8.5 VehicleMotion_SWC Vectoring Algorithm

This section documents the formal mathematical formulations, internal state machine definitions, and execution invariants governing VehicleMotion_SWC Vectoring Algorithm. All equations are executed cyclically by the component's internal runnable entities.

Detailed Algorithmic Specification

Torque request calculation combines driver pedal mapping, cruise control PID output, and speed limit governor commands: T_raw = (Pedal_Ratio * T_max_curve(RPM)) + T_cruise_delta. The result is clipped against the battery power envelope: T_clipped = min(T_raw, (P_battery_discharge_limit / Motor_Angular_Velocity)). When friction braking is active, regenerative torque is blended smoothly: T_final = T_clipped - T_regen_target.

Algorithmic Step	Execution Frequency	Expected Output Range
Step 1: Input Conditioning	10 ms	0.0% to 100.0%
Step 2: Base Torque Lookup	10 ms	-150 Nm to +730 Nm
Step 3: Power Limit Clamp	10 ms	-250 Nm to +730 Nm
Step 4: Stability Adjustment	10 ms	Front: 0-280Nm, Rear: 0-450Nm
Step 5: Safety Verification	2 ms	Boolean Invariant Check

Safety Invariant Verification

Any violation of Step 5 where commanded torque exceeds actual motor torque capability by more than 50 Nm for longer than 20 ms triggers immediate ASIL-D safe stop initiation.

8.6 EmergencyCutoff_SWC Plausibility Logic

This section documents the formal mathematical formulations, internal state machine definitions, and execution invariants governing EmergencyCutoff_SWC Plausibility Logic. All equations are executed cyclically by the component's internal runnable entities.

Detailed Algorithmic Specification

Torque request calculation combines driver pedal mapping, cruise control PID output, and speed limit governor commands: T_raw = (Pedal_Ratio * T_max_curve(RPM)) + T_cruise_delta. The result is clipped against the battery power envelope: T_clipped = min(T_raw, (P_battery_discharge_limit / Motor_Angular_Velocity)). When friction braking is active, regenerative torque is blended smoothly: T_final = T_clipped - T_regen_target.

Algorithmic Step	Execution Frequency	Expected Output Range
Step 1: Input Conditioning	10 ms	0.0% to 100.0%
Step 2: Base Torque Lookup	10 ms	-150 Nm to +730 Nm
Step 3: Power Limit Clamp	10 ms	-250 Nm to +730 Nm
Step 4: Stability Adjustment	10 ms	Front: 0-280Nm, Rear: 0-450Nm
Step 5: Safety Verification	2 ms	Boolean Invariant Check

Safety Invariant Verification

Any violation of Step 5 where commanded torque exceeds actual motor torque capability by more than 50 Nm for longer than 20 ms triggers immediate ASIL-D safe stop initiation.

9.1 Adaptive Execution Management Overview

Execution Management (ara::exec) is responsible for the structured startup, shutdown, deterministic execution, and operational state supervision of all software processes within the Adaptive Platform.

Process Identifier	Binary Executable Path	Startup Order Phase	Function Group Binding	Recovery Action on Crash
Proc_PowertrainMgr		Phase 3 (Applications)	MachineState::Running	
Proc_TorqueArbiter		Phase 3 (Applications)	MachineState::Running	
Proc_BatteryMgr		Phase 2 (Device Services)	MachineState::Running	
Proc_ThermalMgr		Phase 3 (Applications)	MachineState::Running	
Proc_VehicleMotion		Phase 3 (Applications)	MachineState::Running	Fallback to 50:50 Fixed Split
Proc_EmergencyCutoff		Phase 1 (Safety Core)	Always Active	Hardware System Reset

9.2 Process Startup Sequencing & Dependencies

Execution Management (ara::exec) is responsible for the structured startup, shutdown, deterministic execution, and operational state supervision of all software processes within the Adaptive Platform.

Process Identifier	Binary Executable Path	Startup Order Phase	Function Group Binding	Recovery Action on Crash
Proc_PowertrainMgr		Phase 3 (Applications)	MachineState::Running	
Proc_TorqueArbiter		Phase 3 (Applications)	MachineState::Running	
Proc_BatteryMgr		Phase 2 (Device Services)	MachineState::Running	
Proc_ThermalMgr		Phase 3 (Applications)	MachineState::Running	
Proc_VehicleMotion		Phase 3 (Applications)	MachineState::Running	Fallback to 50:50 Fixed Split
Proc_EmergencyCutoff		Phase 1 (Safety Core)	Always Active	Hardware System Reset

9.3 State Management & Function Groups

Execution Management (ara::exec) is responsible for the structured startup, shutdown, deterministic execution, and operational state supervision of all software processes within the Adaptive Platform.

Process Identifier	Binary Executable Path	Startup Order Phase	Function Group Binding	Recovery Action on Crash
Proc_PowertrainMgr		Phase 3 (Applications)	MachineState::Running	
Proc_TorqueArbiter		Phase 3 (Applications)	MachineState::Running	
Proc_BatteryMgr		Phase 2 (Device Services)	MachineState::Running	
Proc_ThermalMgr		Phase 3 (Applications)	MachineState::Running	
Proc_VehicleMotion		Phase 3 (Applications)	MachineState::Running	Fallback to 50:50 Fixed Split
Proc_EmergencyCutoff		Phase 1 (Safety Core)	Always Active	Hardware System Reset

9.4 Execution Client API & Process Health Reporting

Execution Management (ara::exec) is responsible for the structured startup, shutdown, deterministic execution, and operational state supervision of all software processes within the Adaptive Platform.

Process Identifier	Binary Executable Path	Startup Order Phase	Function Group Binding	Recovery Action on Crash
Proc_PowertrainMgr		Phase 3 (Applications)	MachineState::Running	
Proc_TorqueArbiter		Phase 3 (Applications)	MachineState::Running	
Proc_BatteryMgr		Phase 2 (Device Services)	MachineState::Running	
Proc_ThermalMgr		Phase 3 (Applications)	MachineState::Running	
Proc_VehicleMotion		Phase 3 (Applications)	MachineState::Running	Fallback to 50:50 Fixed Split
Proc_EmergencyCutoff		Phase 1 (Safety Core)	Always Active	Hardware System Reset

9.5 Process Failure Containment & Core Recovery

Execution Management (ara::exec) is responsible for the structured startup, shutdown, deterministic execution, and operational state supervision of all software processes within the Adaptive Platform.

Process Identifier	Binary Executable Path	Startup Order Phase	Function Group Binding	Recovery Action on Crash
Proc_PowertrainMgr		Phase 3 (Applications)	MachineState::Running	
Proc_TorqueArbiter		Phase 3 (Applications)	MachineState::Running	
Proc_BatteryMgr		Phase 2 (Device Services)	MachineState::Running	
Proc_ThermalMgr		Phase 3 (Applications)	MachineState::Running	
Proc_VehicleMotion		Phase 3 (Applications)	MachineState::Running	Fallback to 50:50 Fixed Split
Proc_EmergencyCutoff		Phase 1 (Safety Core)	Always Active	Hardware System Reset

10.1 Runnable Catalog: High-Frequency 1ms & 2ms Tasks

Every Adaptive software component organizes its internal computation into discrete runnable entities triggered by periodic timers or asynchronous events.

Runnable Entity Symbol		Trigger Type & Period	SCHED_FIFO Priority	Target Core Affinity
	EmergencyCutoff_SWC	Periodic 1.0 ms	Priority 99 (Max)	Core 3 (Isolated)
	InverterController_SWC	Periodic 2.0 ms	Priority 90	Core 1
	BatteryManagement_SWC	Periodic 5.0 ms	Priority 85	Core 0
	TorqueArbitration_SWC	Periodic 10.0 ms	Priority 80	Core 1
	PowertrainManager_SWC	Periodic 10.0 ms	Priority 75	Core 0
	VehicleMotion_SWC	Periodic 10.0 ms	Priority 70	Core 1
	ThermalManagement_SWC	Periodic 100.0 ms	Priority 40	Core 2
	DiagnosticManager	Periodic 1000.0 ms	Priority 10 (Lowest)	Core 2

10.2 Runnable Catalog: Real-Time 5ms & 10ms Tasks

Every Adaptive software component organizes its internal computation into discrete runnable entities triggered by periodic timers or asynchronous events.

Runnable Entity Symbol		Trigger Type & Period	SCHED_FIFO Priority	Target Core Affinity
	EmergencyCutoff_SWC	Periodic 1.0 ms	Priority 99 (Max)	Core 3 (Isolated)
	InverterController_SWC	Periodic 2.0 ms	Priority 90	Core 1
	BatteryManagement_SWC	Periodic 5.0 ms	Priority 85	Core 0
	TorqueArbitration_SWC	Periodic 10.0 ms	Priority 80	Core 1
	PowertrainManager_SWC	Periodic 10.0 ms	Priority 75	Core 0
	VehicleMotion_SWC	Periodic 10.0 ms	Priority 70	Core 1
	ThermalManagement_SWC	Periodic 100.0 ms	Priority 40	Core 2
	DiagnosticManager	Periodic 1000.0 ms	Priority 10 (Lowest)	Core 2

10.3 Runnable Catalog: Mid-Frequency 20ms & 50ms Tasks

Every Adaptive software component organizes its internal computation into discrete runnable entities triggered by periodic timers or asynchronous events.

Runnable Entity Symbol		Trigger Type & Period	SCHED_FIFO Priority	Target Core Affinity
	EmergencyCutoff_SWC	Periodic 1.0 ms	Priority 99 (Max)	Core 3 (Isolated)
	InverterController_SWC	Periodic 2.0 ms	Priority 90	Core 1
	BatteryManagement_SWC	Periodic 5.0 ms	Priority 85	Core 0
	TorqueArbitration_SWC	Periodic 10.0 ms	Priority 80	Core 1
	PowertrainManager_SWC	Periodic 10.0 ms	Priority 75	Core 0
	VehicleMotion_SWC	Periodic 10.0 ms	Priority 70	Core 1
	ThermalManagement_SWC	Periodic 100.0 ms	Priority 40	Core 2
	DiagnosticManager	Periodic 1000.0 ms	Priority 10 (Lowest)	Core 2

10.4 Runnable Catalog: Low-Frequency 100ms & 1000ms Tasks

Every Adaptive software component organizes its internal computation into discrete runnable entities triggered by periodic timers or asynchronous events.

Runnable Entity Symbol	Trigger Type & Period	SCHED_FIFO Priority	Target Core Affinity
EmergencyCutoff_SWC	Periodic 1.0 ms	Priority 99 (Max)	Core 3 (Isolated)
InverterController_SWC	Periodic 2.0 ms	Priority 90	Core 1
BatteryManagement_SWC	Periodic 5.0 ms	Priority 85	Core 0
TorqueArbitration_SWC	Periodic 10.0 ms	Priority 80	Core 1
PowertrainManager_SWC	Periodic 10.0 ms	Priority 75	Core 0
VehicleMotion_SWC	Periodic 10.0 ms	Priority 70	Core 1
ThermalManagement_SWC	Periodic 100.0 ms	Priority 40	Core 2
DiagnosticManager	Periodic 1000.0 ms	Priority 10 (Lowest)	Core 2

10.5 CPU Core Affinity & Scheduling Priority Allocation

Every Adaptive software component organizes its internal computation into discrete runnable entities triggered by periodic timers or asynchronous events.

Runnable Entity Symbol		Trigger Type & Period	SCHED_FIFO Priority	Target Core Affinity
	EmergencyCutoff_SWC	Periodic 1.0 ms	Priority 99 (Max)	Core 3 (Isolated)
	InverterController_SWC	Periodic 2.0 ms	Priority 90	Core 1
	BatteryManagement_SWC	Periodic 5.0 ms	Priority 85	Core 0
	TorqueArbitration_SWC	Periodic 10.0 ms	Priority 80	Core 1
	PowertrainManager_SWC	Periodic 10.0 ms	Priority 75	Core 0
	VehicleMotion_SWC	Periodic 10.0 ms	Priority 70	Core 1
	ThermalManagement_SWC	Periodic 100.0 ms	Priority 40	Core 2
	DiagnosticManager	Periodic 1000.0 ms	Priority 10 (Lowest)	Core 2

11.1 Service-Oriented Architecture (SOA) Foundations

Communication Management (ara::com) provides a unified, location-transparent communication abstraction for intra-machine IPC and inter-ECU SOME/IP network traffic.

Service Interface Symbol	Service ID	Communication Pattern	Transport Binding	E2E Protection Profile
IPowertrainCoordination	0x1001		SOME/IP over UDP	E2E Profile 11 (CRC-32)
ITorqueCommand	0x1002	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IBatteryStatus	0x1003	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 4 (CRC-16)
IThermalTelemetry	0x1004	Publish-Subscribe Events	SOME/IP over TCP	None (Standard CRC)
IVehicleDynamics	0x1005	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IEmergencySafetyCutoff	0x1000	Direct IPC / Shared Memory	ara::com Zero-Copy	

11.2 Publish-Subscribe Event Communication Patterns

Communication Management (ara::com) provides a unified, location-transparent communication abstraction for intra-machine IPC and inter-ECU SOME/IP network traffic.

Service Interface Symbol	Service ID	Communication Pattern	Transport Binding	E2E Protection Profile
IPowertrainCoordination	0x1001		SOME/IP over UDP	E2E Profile 11 (CRC-32)
ITorqueCommand	0x1002	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IBatteryStatus	0x1003	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 4 (CRC-16)
IThermalTelemetry	0x1004	Publish-Subscribe Events	SOME/IP over TCP	None (Standard CRC)
IVehicleDynamics	0x1005	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IEmergencySafetyCutoff	0x1000	Direct IPC / Shared Memory	ara::com Zero-Copy	

11.3 Request-Response Method RPC Architectures

Communication Management (ara::com) provides a unified, location-transparent communication abstraction for intra-machine IPC and inter-ECU SOME/IP network traffic.

Service Interface Symbol	Service ID	Communication Pattern	Transport Binding	E2E Protection Profile
IPowertrainCoordination	0x1001		SOME/IP over UDP	E2E Profile 11 (CRC-32)
ITorqueCommand	0x1002	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IBatteryStatus	0x1003	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 4 (CRC-16)
IThermalTelemetry	0x1004	Publish-Subscribe Events	SOME/IP over TCP	None (Standard CRC)
IVehicleDynamics	0x1005	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IEmergencySafetyCutoff	0x1000	Direct IPC / Shared Memory	ara::com Zero-Copy	

11.4 End-to-End (E2E) Communication Protection

Communication Management (ara::com) provides a unified, location-transparent communication abstraction for intra-machine IPC and inter-ECU SOME/IP network traffic.

Service Interface Symbol	Service ID	Communication Pattern	Transport Binding	E2E Protection Profile
IPowertrainCoordination	0x1001		SOME/IP over UDP	E2E Profile 11 (CRC-32)
ITorqueCommand	0x1002	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IBatteryStatus	0x1003	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 4 (CRC-16)
IThermalTelemetry	0x1004	Publish-Subscribe Events	SOME/IP over TCP	None (Standard CRC)
IVehicleDynamics	0x1005	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IEmergencySafetyCutoff	0x1000	Direct IPC / Shared Memory	ara::com Zero-Copy	

11.5 Network Serialization & Zero-Copy IPC Optimization

Communication Management (ara::com) provides a unified, location-transparent communication abstraction for intra-machine IPC and inter-ECU SOME/IP network traffic.

Service Interface Symbol	Service ID	Communication Pattern	Transport Binding	E2E Protection Profile
IPowertrainCoordination	0x1001		SOME/IP over UDP	E2E Profile 11 (CRC-32)
ITorqueCommand	0x1002	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IBatteryStatus	0x1003	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 4 (CRC-16)
IThermalTelemetry	0x1004	Publish-Subscribe Events	SOME/IP over TCP	None (Standard CRC)
IVehicleDynamics	0x1005	Publish-Subscribe Events	SOME/IP over UDP	E2E Profile 11 (CRC-32)
IEmergencySafetyCutoff	0x1000	Direct IPC / Shared Memory	ara::com Zero-Copy	

12.1 Service Interface: IPowertrainCoordination (0x1001)

This section formalizes the exact data structures, member types, scaling equations, physical boundaries, and invalid value indicators for this interface.

Field / Member Name	Data Type	Scaling Factor	Physical Range	Invalid Value Indicator
TargetTorque_Front	int16_t	0.1 Nm / LSB	-300.0 to +300.0 Nm	0x7FFF (Invalid)
TargetTorque_Rear	int16_t	0.1 Nm / LSB	-500.0 to +500.0 Nm	0x7FFF (Invalid)
TorqueLimit_Discharge	uint16_t	0.1 Nm / LSB	0.0 to +800.0 Nm	0xFFFF (Invalid)
TorqueLimit_Regen	uint16_t	0.1 Nm / LSB	0.0 to +400.0 Nm	0xFFFF (Invalid)
SequenceCounter	uint8_t	1 unit / LSB	0 to 255 (Rolling)	No Invalid State
SafetyCRC	uint32_t	Bitmask	0x00000000 - 0xFFFFFFFF	0x00000000

12.2 Service Interface: ITorqueCommand (0x1002)

This section formalizes the exact data structures, member types, scaling equations, physical boundaries, and invalid value indicators for this interface.

Field / Member Name	Data Type	Scaling Factor	Physical Range	Invalid Value Indicator
TargetTorque_Front	int16_t	0.1 Nm / LSB	-300.0 to +300.0 Nm	0x7FFF (Invalid)
TargetTorque_Rear	int16_t	0.1 Nm / LSB	-500.0 to +500.0 Nm	0x7FFF (Invalid)
TorqueLimit_Discharge	uint16_t	0.1 Nm / LSB	0.0 to +800.0 Nm	0xFFFF (Invalid)
TorqueLimit_Regen	uint16_t	0.1 Nm / LSB	0.0 to +400.0 Nm	0xFFFF (Invalid)
SequenceCounter	uint8_t	1 unit / LSB	0 to 255 (Rolling)	No Invalid State
SafetyCRC	uint32_t	Bitmask	0x00000000 - 0xFFFFFFFF	0x00000000

12.3 Service Interface: IBatteryStatus (0x1003)

This section formalizes the exact data structures, member types, scaling equations, physical boundaries, and invalid value indicators for this interface.

Field / Member Name	Data Type	Scaling Factor	Physical Range	Invalid Value Indicator
TargetTorque_Front	int16_t	0.1 Nm / LSB	-300.0 to +300.0 Nm	0x7FFF (Invalid)
TargetTorque_Rear	int16_t	0.1 Nm / LSB	-500.0 to +500.0 Nm	0x7FFF (Invalid)
TorqueLimit_Discharge	uint16_t	0.1 Nm / LSB	0.0 to +800.0 Nm	0xFFFF (Invalid)
TorqueLimit_Regen	uint16_t	0.1 Nm / LSB	0.0 to +400.0 Nm	0xFFFF (Invalid)
SequenceCounter	uint8_t	1 unit / LSB	0 to 255 (Rolling)	No Invalid State
SafetyCRC	uint32_t	Bitmask	0x00000000 - 0xFFFFFFFF	0x00000000

12.4 Service Interface: IThermalTelemetry (0x1004)

This section formalizes the exact data structures, member types, scaling equations, physical boundaries, and invalid value indicators for this interface.

Field / Member Name	Data Type	Scaling Factor	Physical Range	Invalid Value Indicator
TargetTorque_Front	int16_t	0.1 Nm / LSB	-300.0 to +300.0 Nm	0x7FFF (Invalid)
TargetTorque_Rear	int16_t	0.1 Nm / LSB	-500.0 to +500.0 Nm	0x7FFF (Invalid)
TorqueLimit_Discharge	uint16_t	0.1 Nm / LSB	0.0 to +800.0 Nm	0xFFFF (Invalid)
TorqueLimit_Regen	uint16_t	0.1 Nm / LSB	0.0 to +400.0 Nm	0xFFFF (Invalid)
SequenceCounter	uint8_t	1 unit / LSB	0 to 255 (Rolling)	No Invalid State
SafetyCRC	uint32_t	Bitmask	0x00000000 - 0xFFFFFFFF	0x00000000

12.5 Service Interface: IVehicleDynamics (0x1005)

This section formalizes the exact data structures, member types, scaling equations, physical boundaries, and invalid value indicators for this interface.

Field / Member Name	Data Type	Scaling Factor	Physical Range	Invalid Value Indicator
TargetTorque_Front	int16_t	0.1 Nm / LSB	-300.0 to +300.0 Nm	0x7FFF (Invalid)
TargetTorque_Rear	int16_t	0.1 Nm / LSB	-500.0 to +500.0 Nm	0x7FFF (Invalid)
TorqueLimit_Discharge	uint16_t	0.1 Nm / LSB	0.0 to +800.0 Nm	0xFFFF (Invalid)
TorqueLimit_Regen	uint16_t	0.1 Nm / LSB	0.0 to +400.0 Nm	0xFFFF (Invalid)
SequenceCounter	uint8_t	1 unit / LSB	0 to 255 (Rolling)	No Invalid State
SafetyCRC	uint32_t	Bitmask	0x00000000 - 0xFFFFFFFF	0x00000000

12.6 Service Interface: IEmergencySafetyCutoff (0x1000)

This section formalizes the exact data structures, member types, scaling equations, physical boundaries, and invalid value indicators for this interface.

Field / Member Name	Data Type	Scaling Factor	Physical Range	Invalid Value Indicator
TargetTorque_Front	int16_t	0.1 Nm / LSB	-300.0 to +300.0 Nm	0x7FFF (Invalid)
TargetTorque_Rear	int16_t	0.1 Nm / LSB	-500.0 to +500.0 Nm	0x7FFF (Invalid)
TorqueLimit_Discharge	uint16_t	0.1 Nm / LSB	0.0 to +800.0 Nm	0xFFFF (Invalid)
TorqueLimit_Regen	uint16_t	0.1 Nm / LSB	0.0 to +400.0 Nm	0xFFFF (Invalid)
SequenceCounter	uint8_t	1 unit / LSB	0 to 255 (Rolling)	No Invalid State
SafetyCRC	uint32_t	Bitmask	0x00000000 - 0xFFFFFFFF	0x00000000

12.7 Complete Signal Dictionary & Physical Unit Mapping

This section formalizes the exact data structures, member types, scaling equations, physical boundaries, and invalid value indicators for this interface.

Field / Member Name	Data Type	Scaling Factor	Physical Range	Invalid Value Indicator
TargetTorque_Front	int16_t	0.1 Nm / LSB	-300.0 to +300.0 Nm	0x7FFF (Invalid)
TargetTorque_Rear	int16_t	0.1 Nm / LSB	-500.0 to +500.0 Nm	0x7FFF (Invalid)
TorqueLimit_Discharge	uint16_t	0.1 Nm / LSB	0.0 to +800.0 Nm	0xFFFF (Invalid)
TorqueLimit_Regen	uint16_t	0.1 Nm / LSB	0.0 to +400.0 Nm	0xFFFF (Invalid)
SequenceCounter	uint8_t	1 unit / LSB	0 to 255 (Rolling)	No Invalid State
SafetyCRC	uint32_t	Bitmask	0x00000000 - 0xFFFFFFFF	0x00000000

13.1 SOME/IP Header Format & Serialization Standard

Scalable service-Oriented MiddlewarE over IP (SOME/IP) is the automotive standard transport protocol for high-bandwidth Adaptive AUTOSAR communication.

Network Parameter	Configured Value	Protocol Reference	Network Architecture Impact
ePDC IP Address	192.168.10.15 / 24	IPv4 Static Host	
SOME/IP-SD Multicast IP	239.255.0.1	IETF RFC 2365	
SOME/IP-SD UDP Port	30490	AUTOSAR SD Specification	Standardized service discovery port
Real-Time VLAN ID	VLAN 100 (Priority 6)	IEEE 802.1Q	
Diagnostic VLAN ID	VLAN 200 (Priority 2)	IEEE 802.1Q	

13.2 Service Discovery (SOME/IP-SD) Entry Formats

Scalable service-Oriented MiddlewarE over IP (SOME/IP) is the automotive standard transport protocol for high-bandwidth Adaptive AUTOSAR communication.

Network Parameter	Configured Value	Protocol Reference	Network Architecture Impact
ePDC IP Address	192.168.10.15 / 24	IPv4 Static Host	
SOME/IP-SD Multicast IP	239.255.0.1	IETF RFC 2365	
SOME/IP-SD UDP Port	30490	AUTOSAR SD Specification	Standardized service discovery port
Real-Time VLAN ID	VLAN 100 (Priority 6)	IEEE 802.1Q	
Diagnostic VLAN ID	VLAN 200 (Priority 2)	IEEE 802.1Q	

13.3 Network Topology: IP Endpoints & VLAN Partitioning

Scalable service-Oriented MiddlewarE over IP (SOME/IP) is the automotive standard transport protocol for high-bandwidth Adaptive AUTOSAR communication.

Network Parameter	Configured Value	Protocol Reference	Network Architecture Impact
ePDC IP Address	192.168.10.15 / 24	IPv4 Static Host	
SOME/IP-SD Multicast IP	239.255.0.1	IETF RFC 2365	
SOME/IP-SD UDP Port	30490	AUTOSAR SD Specification	Standardized service discovery port
Real-Time VLAN ID	VLAN 100 (Priority 6)	IEEE 802.1Q	
Diagnostic VLAN ID	VLAN 200 (Priority 2)	IEEE 802.1Q	

13.4 Eventgroup Subscription & Multicast Routing

Scalable service-Oriented MiddlewarE over IP (SOME/IP) is the automotive standard transport protocol for high-bandwidth Adaptive AUTOSAR communication.

Network Parameter	Configured Value	Protocol Reference	Network Architecture Impact
ePDC IP Address	192.168.10.15 / 24	IPv4 Static Host	
SOME/IP-SD Multicast IP	239.255.0.1	IETF RFC 2365	
SOME/IP-SD UDP Port	30490	AUTOSAR SD Specification	Standardized service discovery port
Real-Time VLAN ID	VLAN 100 (Priority 6)	IEEE 802.1Q	
Diagnostic VLAN ID	VLAN 200 (Priority 2)	IEEE 802.1Q	

13.5 SOME/IP Error Codes & RPC Return Codes

Scalable service-Oriented MiddlewarE over IP (SOME/IP) is the automotive standard transport protocol for high-bandwidth Adaptive AUTOSAR communication.

Network Parameter	Configured Value	Protocol Reference	Network Architecture Impact
ePDC IP Address	192.168.10.15 / 24	IPv4 Static Host	
SOME/IP-SD Multicast IP	239.255.0.1	IETF RFC 2365	
SOME/IP-SD UDP Port	30490	AUTOSAR SD Specification	Standardized service discovery port
Real-Time VLAN ID	VLAN 100 (Priority 6)	IEEE 802.1Q	
Diagnostic VLAN ID	VLAN 200 (Priority 2)	IEEE 802.1Q	

14.1 Persistency Management (ara::per) Architecture

Persistency Management (ara::per) guarantees deterministic, atomic access to non-volatile flash storage across vehicle power loss events.

Persisted Item Key	Data Structure Type	Max Size	Update Frequency	Integrity Protection
CFG_PedalResponseMap	Array of Float (20x2)	320 Bytes	Manufacturing / Service	CRC-32 + Digital Signature
CAL_MotorThermalLimits	Struct ThermalCoeffs	128 Bytes	Manufacturing / Service	CRC-32
NV_OdometerTotalEnergy	uint64_t (Watt-hours)	8 Bytes	Every 1.0 km traveled	Double-buffered + Journal
NV_ContactorCycleCount	uint32_t (Cycles)	4 Bytes	On vehicle shutdown	Atomic Flash Write
DTC_FreezeFrameStore	Array of Snapshots (64)	16.0 KB	On diagnostic event	Crash-safe File Storage

14.2 Non-Volatile Calibration Parameter Catalog

Persistency Management (ara::per) guarantees deterministic, atomic access to non-volatile flash storage across vehicle power loss events.

Persisted Item Key	Data Structure Type	Max Size	Update Frequency	Integrity Protection
CFG_PedalResponseMap	Array of Float (20x2)	320 Bytes	Manufacturing / Service	CRC-32 + Digital Signature
CAL_MotorThermalLimits	Struct ThermalCoeffs	128 Bytes	Manufacturing / Service	CRC-32
NV_OdometerTotalEnergy	uint64_t (Watt-hours)	8 Bytes	Every 1.0 km traveled	Double-buffered + Journal
NV_ContactorCycleCount	uint32_t (Cycles)	4 Bytes	On vehicle shutdown	Atomic Flash Write
DTC_FreezeFrameStore	Array of Snapshots (64)	16.0 KB	On diagnostic event	Crash-safe File Storage

14.3 Flash Wear-Leveling & Redundancy Pools

Persistency Management (ara::per) guarantees deterministic, atomic access to non-volatile flash storage across vehicle power loss events.

Persisted Item Key	Data Structure Type	Max Size	Update Frequency	Integrity Protection
CFG_PedalResponseMap	Array of Float (20x2)	320 Bytes	Manufacturing / Service	CRC-32 + Digital Signature
CAL_MotorThermalLimits	Struct ThermalCoeffs	128 Bytes	Manufacturing / Service	CRC-32
NV_OdometerTotalEnergy	uint64_t (Watt-hours)	8 Bytes	Every 1.0 km traveled	Double-buffered + Journal
NV_ContactorCycleCount	uint32_t (Cycles)	4 Bytes	On vehicle shutdown	Atomic Flash Write
DTC_FreezeFrameStore	Array of Snapshots (64)	16.0 KB	On diagnostic event	Crash-safe File Storage

14.4 Crash-Safe Transactional Persistence Strategy

Persistency Management (ara::per) guarantees deterministic, atomic access to non-volatile flash storage across vehicle power loss events.

Persisted Item Key	Data Structure Type	Max Size	Update Frequency	Integrity Protection
CFG_PedalResponseMap	Array of Float (20x2)	320 Bytes	Manufacturing / Service	CRC-32 + Digital Signature
CAL_MotorThermalLimits	Struct ThermalCoeffs	128 Bytes	Manufacturing / Service	CRC-32
NV_OdometerTotalEnergy	uint64_t (Watt-hours)	8 Bytes	Every 1.0 km traveled	Double-buffered + Journal
NV_ContactorCycleCount	uint32_t (Cycles)	4 Bytes	On vehicle shutdown	Atomic Flash Write
DTC_FreezeFrameStore	Array of Snapshots (64)	16.0 KB	On diagnostic event	Crash-safe File Storage

15.1 Diagnostic Management (ara::diag) & DoIP Architecture

Diagnostic Management (ara::diag) provides standardized UDS service routing over Ethernet DoIP (ISO 13400) for offboard scan tools.

DTC Identifier	Fault Trigger Condition	Debounce Filter	Primary System Action
DTC_P0A1F		Immediate (0ms)	
DTC_P0A80		10 drive cycles	
DTC_P0C00		20 ms continuous	
DTC_U0100		100 ms timeout	
DTC_P0A0D	HVIL continuity signal lost	10 ms immediate	

15.2 Diagnostic Trouble Code (DTC) Master Catalog

Diagnostic Management (ara::diag) provides standardized UDS service routing over Ethernet DoIP (ISO 13400) for offboard scan tools.

DTC Identifier	Fault Trigger Condition	Debounce Filter	Primary System Action
DTC_P0A1F		Immediate (0ms)	
DTC_P0A80		10 drive cycles	
DTC_P0C00		20 ms continuous	
DTC_U0100		100 ms timeout	
DTC_P0A0D	HVIL continuity signal lost	10 ms immediate	

15.3 Freeze Frame Data Definitions & Trigger Conditions

Diagnostic Management (ara::diag) provides standardized UDS service routing over Ethernet DoIP (ISO 13400) for offboard scan tools.

DTC Identifier	Fault Trigger Condition	Debounce Filter	Primary System Action
DTC_P0A1F		Immediate (0ms)	
DTC_P0A80		10 drive cycles	
DTC_P0C00		20 ms continuous	
DTC_U0100		100 ms timeout	
DTC_P0A0D	HVIL continuity signal lost	10 ms immediate	

15.4 Diagnostic Log and Trace (DLT) Logging Framework

Diagnostic Management (ara::diag) provides standardized UDS service routing over Ethernet DoIP (ISO 13400) for offboard scan tools.

DTC Identifier	Fault Trigger Condition	Debounce Filter	Primary System Action
DTC_P0A1F		Immediate (0ms)	
DTC_P0A80		10 drive cycles	
DTC_P0C00		20 ms continuous	
DTC_U0100		100 ms timeout	
DTC_P0A0D	HVIL continuity signal lost	10 ms immediate	

15.5 Supervised Entity Health Monitoring & Watchdog Integration

Diagnostic Management (ara::diag) provides standardized UDS service routing over Ethernet DoIP (ISO 13400) for offboard scan tools.

DTC Identifier	Fault Trigger Condition	Debounce Filter	Primary System Action
DTC_P0A1F		Immediate (0ms)	
DTC_P0A80		10 drive cycles	
DTC_P0C00		20 ms continuous	
DTC_U0100		100 ms timeout	
DTC_P0A0D	HVIL continuity signal lost	10 ms immediate	

16.1 Fault Containment Regions (FCR) Architecture

The powertrain architecture implements layered Fault Containment Regions (FCRs) to isolate component failures before they propagate into hazard states.

Degraded Mode Identifier	Activation Trigger	Torque Capability Limit	Vehicle Speed Cap	Driver Warning
LIMP_INVERTER_FAIL			80 km/h	Amber Powertrain Warning
LIMP_BATTERY_HOT			100 km/h	
LIMP_COOLING_PUMP			60 km/h	
SAFE_STOP_TORQUE			Coast to stop	

16.2 Limp-Home Degraded Operating Modes

The powertrain architecture implements layered Fault Containment Regions (FCRs) to isolate component failures before they propagate into hazard states.

Degraded Mode Identifier	Activation Trigger	Torque Capability Limit	Vehicle Speed Cap	Driver Warning
LIMP_INVERTER_FAIL			80 km/h	Amber Powertrain Warning
LIMP_BATTERY_HOT			100 km/h	
LIMP_COOLING_PUMP			60 km/h	
SAFE_STOP_TORQUE			Coast to stop	

16.3 Emergency Contactor Opening & Pyrofuse Deployment

The powertrain architecture implements layered Fault Containment Regions (FCRs) to isolate component failures before they propagate into hazard states.

Degraded Mode Identifier	Activation Trigger	Torque Capability Limit	Vehicle Speed Cap	Driver Warning
LIMP_INVERTER_FAIL			80 km/h	Amber Powertrain Warning
LIMP_BATTERY_HOT			100 km/h	
LIMP_COOLING_PUMP			60 km/h	
SAFE_STOP_TORQUE			Coast to stop	

16.4 Thermal Derating Envelopes & Power Throttling

The powertrain architecture implements layered Fault Containment Regions (FCRs) to isolate component failures before they propagate into hazard states.

Degraded Mode Identifier	Activation Trigger	Torque Capability Limit	Vehicle Speed Cap	Driver Warning
LIMP_INVERTER_FAIL			80 km/h	Amber Powertrain Warning
LIMP_BATTERY_HOT			100 km/h	
LIMP_COOLING_PUMP			60 km/h	
SAFE_STOP_TORQUE			Coast to stop	

17.1 Item Definition & Hazard Analysis and Risk Assessment (HARA)

In accordance with ISO 26262:2018, this chapter formalizes the functional safety concept and safety goals protecting vehicle occupants from powertrain hazards.

Hazard ID	Severity (S)	Exposure (E)	Controllability (C)	Safety Goal & ASIL
HAZ-001	S3 (Life threatening)	E4 (High exposure)	C3 (Difficult to control)	
HAZ-002	S3 (Life threatening)		C3 (Difficult to control)	
HAZ-003	S3 (Fatal shock)	E2 (Low exposure)	C3 (Uncontrollable)	
HAZ-004	S3 (Fire / Explosion)		C2 (Partial control)	

17.2 Functional Safety Concept (FSC) & Safety Goals

In accordance with ISO 26262:2018, this chapter formalizes the functional safety concept and safety goals protecting vehicle occupants from powertrain hazards.

Hazard ID	Severity (S)	Exposure (E)	Controllability (C)	Safety Goal & ASIL
HAZ-001	S3 (Life threatening)	E4 (High exposure)	C3 (Difficult to control)	
HAZ-002	S3 (Life threatening)		C3 (Difficult to control)	
HAZ-003	S3 (Fatal shock)	E2 (Low exposure)	C3 (Uncontrollable)	
HAZ-004	S3 (Fire / Explosion)		C2 (Partial control)	

17.3 Technical Safety Concept (TSC) & Architecture Decomposition

In accordance with ISO 26262:2018, this chapter formalizes the functional safety concept and safety goals protecting vehicle occupants from powertrain hazards.

Hazard ID	Severity (S)	Exposure (E)	Controllability (C)	Safety Goal & ASIL
HAZ-001	S3 (Life threatening)	E4 (High exposure)	C3 (Difficult to control)	
HAZ-002	S3 (Life threatening)		C3 (Difficult to control)	
HAZ-003	S3 (Fatal shock)	E2 (Low exposure)	C3 (Uncontrollable)	
HAZ-004	S3 (Fire / Explosion)		C2 (Partial control)	

17.4 Hardware-Software Fault Metrics & Diagnostic Coverage

In accordance with ISO 26262:2018, this chapter formalizes the functional safety concept and safety goals protecting vehicle occupants from powertrain hazards.

Hazard ID	Severity (S)	Exposure (E)	Controllability (C)	Safety Goal & ASIL
HAZ-001	S3 (Life threatening)	E4 (High exposure)	C3 (Difficult to control)	
HAZ-002	S3 (Life threatening)		C3 (Difficult to control)	
HAZ-003	S3 (Fatal shock)	E2 (Low exposure)	C3 (Uncontrollable)	
HAZ-004	S3 (Fire / Explosion)		C2 (Partial control)	

17.5 Safety Mechanisms & E2E Protection Implementations

In accordance with ISO 26262:2018, this chapter formalizes the functional safety concept and safety goals protecting vehicle occupants from powertrain hazards.

Hazard ID	Severity (S)	Exposure (E)	Controllability (C)	Safety Goal & ASIL
HAZ-001	S3 (Life threatening)	E4 (High exposure)	C3 (Difficult to control)	
HAZ-002	S3 (Life threatening)		C3 (Difficult to control)	
HAZ-003	S3 (Fatal shock)	E2 (Low exposure)	C3 (Uncontrollable)	
HAZ-004	S3 (Fire / Explosion)		C2 (Partial control)	

18.1 Threat Analysis and Risk Assessment (TARA)

This chapter defines the cybersecurity architecture safeguarding the powertrain against unauthorized remote manipulation in compliance with ISO/SAE 21434 and UNECE R155.

Threat ID	Threat Scenario (STRIDE)	Impact	Attack Feasibility
THR-001		Severe (Crash)	Low (Requires bus access)
THR-002		Severe (ECU takeover)	Moderate (Network attack)
THR-003		Moderate (Data access)	Low (Sandboxed POSIX)
THR-004		Moderate (Comms loss)	Moderate (Ethernet flood)

18.2 Secure Onboard Communication (SecOC) Specifications

This chapter defines the cybersecurity architecture safeguarding the powertrain against unauthorized remote manipulation in compliance with ISO/SAE 21434 and UNECE R155.

Threat ID	Threat Scenario (STRIDE)	Impact	Attack Feasibility
THR-001		Severe (Crash)	Low (Requires bus access)
THR-002		Severe (ECU takeover)	Moderate (Network attack)
THR-003		Moderate (Data access)	Low (Sandboxed POSIX)
THR-004		Moderate (Comms loss)	Moderate (Ethernet flood)

18.3 Transport Layer Security (TLS 1.3) & DoIP Encryption

This chapter defines the cybersecurity architecture safeguarding the powertrain against unauthorized remote manipulation in compliance with ISO/SAE 21434 and UNECE R155.

Threat ID	Threat Scenario (STRIDE)	Impact	Attack Feasibility
THR-001		Severe (Crash)	Low (Requires bus access)
THR-002		Severe (ECU takeover)	Moderate (Network attack)
THR-003		Moderate (Data access)	Low (Sandboxed POSIX)
THR-004		Moderate (Comms loss)	Moderate (Ethernet flood)

18.4 Secure Boot, Crypto Stack & Key Storage Architecture

This chapter defines the cybersecurity architecture safeguarding the powertrain against unauthorized remote manipulation in compliance with ISO/SAE 21434 and UNECE R155.

Threat ID	Threat Scenario (STRIDE)	Impact	Attack Feasibility
THR-001		Severe (Crash)	Low (Requires bus access)
THR-002		Severe (ECU takeover)	Moderate (Network attack)
THR-003		Moderate (Data access)	Low (Sandboxed POSIX)
THR-004		Moderate (Comms loss)	Moderate (Ethernet flood)

19.1 End-to-End Latency Profiles across Vehicle Networks

This chapter documents empirical performance measurements, network traffic shaping budgets, and memory allocation maps.

	VLAN Tag	Allocated Bandwidth	Max Burst Size	Max Queuing Delay Target
	VLAN 100	25.0 Mbps	1500 Bytes	< 0.25 ms
	VLAN 100	10.0 Mbps	1000 Bytes	< 0.50 ms
	VLAN 150	50.0 Mbps	9000 Bytes	< 2.0 ms
	VLAN 200	150.0 Mbps	65535 Bytes	< 10.0 ms
	VLAN 250	50.0 Mbps	32768 Bytes	< 50.0 ms

19.2 Ethernet Bandwidth Allocation & TSN Traffic Classes

This chapter documents empirical performance measurements, network traffic shaping budgets, and memory allocation maps.

	VLAN Tag	Allocated Bandwidth	Max Burst Size	Max Queuing Delay Target
	VLAN 100	25.0 Mbps	1500 Bytes	< 0.25 ms
	VLAN 100	10.0 Mbps	1000 Bytes	< 0.50 ms
	VLAN 150	50.0 Mbps	9000 Bytes	< 2.0 ms
	VLAN 200	150.0 Mbps	65535 Bytes	< 10.0 ms
	VLAN 250	50.0 Mbps	32768 Bytes	< 50.0 ms

19.3 Memory Footprint & Non-Volatile Flash Allocation

This chapter documents empirical performance measurements, network traffic shaping budgets, and memory allocation maps.

	VLAN Tag	Allocated Bandwidth	Max Burst Size	Max Queuing Delay Target
	VLAN 100	25.0 Mbps	1500 Bytes	< 0.25 ms
	VLAN 100	10.0 Mbps	1000 Bytes	< 0.50 ms
	VLAN 150	50.0 Mbps	9000 Bytes	< 2.0 ms
	VLAN 200	150.0 Mbps	65535 Bytes	< 10.0 ms
	VLAN 250	50.0 Mbps	32768 Bytes	< 50.0 ms

20.1 Cold Startup & Hardware Power-On Sequencing

This chapter defines the state sequence governing hardware power-up, fast cold-boot, graceful shutdown, and atomic over-the-air firmware updates via UCM.

Startup Phase	Governing Subsystem	Execution Actions	Max Allowable Duration	Exit Criteria
	Hardware / HSM		25 ms	Signature Valid
Phase 1: OS Kernel Boot	POSIX RTOS		150 ms	POSIX PSE52 Ready
Phase 2: Platform Services	ara::exec		200 ms	Platform Services Up
Phase 3: Applications	ara::exec		150 ms	Ready State Reached
Total Startup Budget	End-to-End System		525 ms (Budget < 600 ms)	RUN_ACTIVE Achievable

20.2 Graceful Shutdown & Fast Wakeup Sleep Sequencing

This chapter defines the state sequence governing hardware power-up, fast cold-boot, graceful shutdown, and atomic over-the-air firmware updates via UCM.

Startup Phase	Governing Subsystem	Execution Actions	Max Allowable Duration	Exit Criteria
	Hardware / HSM		25 ms	Signature Valid
Phase 1: OS Kernel Boot	POSIX RTOS		150 ms	POSIX PSE52 Ready
Phase 2: Platform Services	ara::exec		200 ms	Platform Services Up
Phase 3: Applications	ara::exec		150 ms	Ready State Reached
Total Startup Budget	End-to-End System		525 ms (Budget < 600 ms)	RUN_ACTIVE Achievable

20.3 Update and Configuration Management (UCM) & A/B Partitions

This chapter defines the state sequence governing hardware power-up, fast cold-boot, graceful shutdown, and atomic over-the-air firmware updates via UCM.

Startup Phase	Governing Subsystem	Execution Actions	Max Allowable Duration	Exit Criteria
	Hardware / HSM		25 ms	Signature Valid
Phase 1: OS Kernel Boot	POSIX RTOS		150 ms	POSIX PSE52 Ready
Phase 2: Platform Services	ara::exec		200 ms	Platform Services Up
Phase 3: Applications	ara::exec		150 ms	Ready State Reached
Total Startup Budget	End-to-End System		525 ms (Budget < 600 ms)	RUN_ACTIVE Achievable

21.1 Verification Methodology: MIL, SIL, and HIL Levels

This chapter documents the multi-tier verification and validation methodology confirming that all 40 functional requirements and safety goals are satisfied.

Test Level	Execution Platform	Automation Framework
MIL (Model-in-the-Loop)	Simulink / MATLAB	Simulink Test
SIL (Software-in-the-Loop)	Linux x86 Virtual Machine	Google Test / CTest
HIL (Hardware-in-the-Loop)		Python / AutomationDesk
Vehicle Test Track	Prototype Electric Vehicle	Vector CANoe / ETAS INCA

21.2 Fault Injection Testing & Safety Case Validation

This chapter documents the multi-tier verification and validation methodology confirming that all 40 functional requirements and safety goals are satisfied.

Test Level	Execution Platform	Automation Framework
MIL (Model-in-the-Loop)	Simulink / MATLAB	Simulink Test
SIL (Software-in-the-Loop)	Linux x86 Virtual Machine	Google Test / CTest
HIL (Hardware-in-the-Loop)		Python / AutomationDesk
Vehicle Test Track	Prototype Electric Vehicle	Vector CANoe / ETAS INCA

21.3 System Acceptance Criteria & Homologation Gates

This chapter documents the multi-tier verification and validation methodology confirming that all 40 functional requirements and safety goals are satisfied.

Test Level	Execution Platform	Automation Framework
MIL (Model-in-the-Loop)	Simulink / MATLAB	Simulink Test
SIL (Software-in-the-Loop)	Linux x86 Virtual Machine	Google Test / CTest
HIL (Hardware-in-the-Loop)		Python / AutomationDesk
Vehicle Test Track	Prototype Electric Vehicle	Vector CANoe / ETAS INCA

22.1 Architecture Decision Records (ADRs)

During the design of the Electric Powertrain Domain Controller, several fundamental architectural trade-offs were evaluated. The table below formalizes key design decisions, rejected alternatives, and architectural rationales.

ADR Identifier	Design Decision Made	Rejected Alternative	Primary Engineering Rationale
ADR-001			
ADR-002			
ADR-003		Silicon IGBT standard inverters	
ADR-004			
ADR-005		Single-partition in-place flashing	

22.2 Deliberately Documented Omissions & Out-of-Scope Items

To establish clear functional boundaries for system verification and AI evaluation benchmarks, the following domains and capabilities are EXPLICITLY EXCLUDED and documented as OUT OF SCOPE. The software components described herein contain NO implementation, logic, or interfaces for these topics.

Omission ID	Excluded Domain / Feature	Allocated External Subsystem
OMS-001		Body Domain Controller (BDC)
OMS-002		ADAS Autonomous Driving Core
OMS-003		Infotainment Head Unit (IHU)
OMS-004		Electronic Stability Control (ESC)
OMS-005		Offboard Cloud Fleet Analytics
OMS-006		Electric Power Steering (EPS)
OMS-007		Restraints Control Module (RCM)
OMS-008		Telematics Gateway Module (TGM)
OMS-009		
OMS-010		None (Deprecated in architecture)

Evaluation Benchmark Anchor

Queries regarding cabin air conditioning, camera object recognition, brake caliper hydraulic fluid pressure, or tire pressure RF frequencies MUST result in an honest system abstention confirming that these features are explicitly out of scope.

23.1 Requirements Traceability Matrix (Requirements 001-020)

The Requirements Traceability Matrix establishes bidirectional traceability between system requirements, architecture components, design sections, and verification test cases.

Requirement ID	Requirement Title	Implementing SWC	Design Section	Verification Test Case
REQ-PWR-001		PowertrainManager_SWC	Section 7.1 / 8.1	TC-PWR-HIL-001
REQ-PWR-002		EmergencyCutoff_SWC	Section 7.6 / 8.6	TC-SAF-HIL-002
REQ-PWR-003		VehicleMotion_SWC	Section 7.5 / 8.5	TC-MOT-HIL-003
REQ-PWR-004		PowertrainManager_SWC	Section 7.1 / 8.1	TC-PWR-HIL-004
REQ-PWR-005		TorqueArbitration_SWC	Section 7.2 / 8.2	TC-REG-HIL-005
REQ-PWR-006		TorqueArbitration_SWC	Section 7.2 / 8.2	TC-BAT-HIL-006
REQ-PWR-007		ThermalManagement_SWC	Section 7.4 / 8.4	TC-THM-HIL-007
REQ-PWR-008		BatteryManagement_SWC	Section 7.3 / 8.3	TC-BAT-HIL-008
REQ-PWR-009	Contactor Weld Detection	BatteryManagement_SWC	Section 7.3 / 8.3	TC-BAT-HIL-009
REQ-PWR-010		EmergencyCutoff_SWC	Section 7.6 / 8.6	TC-SAF-HIL-010
REQ-PWR-011	Crash Signal Power Cutoff	EmergencyCutoff_SWC	Section 7.6 / 8.6	TC-SAF-HIL-011
REQ-PWR-012		ThermalManagement_SWC	Section 7.4 / 8.4	TC-THM-HIL-012
REQ-PWR-013		InverterController_SWC	Section 7.2 / 8.2	TC-INV-HIL-013
REQ-PWR-014	Motor Stall Protection	TorqueArbitration_SWC	Section 7.2 / 8.2	TC-PWR-HIL-014

23.2 Requirements Traceability Matrix (Requirements 021-040)

Continuation of the Requirements Traceability Matrix connecting requirements 021 through 040 to software components, implementation sections, and test suites.

Requirement ID	Requirement Title	Implementing SWC	Design Section	Verification Test Case
REQ-PWR-021		BatteryManagement_SWC	Section 7.3 / 8.3	TC-BAT-HIL-021
REQ-PWR-022		BatteryManagement_SWC	Section 7.3 / 8.3	TC-BAT-HIL-022
REQ-PWR-023		ThermalManagement_SWC	Section 7.4 / 8.4	TC-THM-HIL-023
REQ-PWR-024		RegenerativeBraking_SWC	Section 7.2 / 8.2	TC-REG-HIL-024
REQ-PWR-025	Creep Torque Control	TorqueArbitration_SWC	Section 7.2 / 8.2	TC-PWR-HIL-025
REQ-PWR-026		BatteryManagement_SWC	Section 7.3 / 8.3	TC-BAT-HIL-026
REQ-PWR-027		EmergencyCutoff_SWC	Section 7.6 / 8.6	TC-SAF-HIL-027
REQ-PWR-028		TorqueArbitration_SWC	Section 7.2 / 8.2	TC-PWR-HIL-028
REQ-PWR-029		InverterController_SWC	Section 7.2 / 8.2	TC-INV-HIL-029
REQ-PWR-030		TorqueArbitration_SWC	Section 11.4 / 13.1	TC-COM-HIL-030
REQ-PWR-035		PowertrainManager_SWC	Section 7.1 / 8.1	TC-PWR-HIL-035
REQ-PWR-038		TorqueArbitration_SWC	Section 7.2 / 8.2	TC-PWR-HIL-038
REQ-PWR-039		PowertrainManager_SWC	Section 16.2	TC-ERR-HIL-039
REQ-PWR-040		EmergencyCutoff_SWC	Section 15.5	TC-SAF-HIL-040

23.3 Technical Glossary & Acronym Dictionary

This glossary defines standard terminology, acronyms, and automotive architectural abbreviations utilized throughout this specification.

Acronym / Term	Full Engineering Expansion	Standardizing Body	Architectural Definition in ePDC
ADAS		SAE J3016	
ARA	AUTOSAR Runtime for Applications	AUTOSAR Adaptive	
ASIL	Automotive Safety Integrity Level	ISO 26262	
AVAS	Acoustic Vehicle Alerting System	UNECE R138	
BMS	Battery Management System	Automotive Standard	
DoIP	Diagnostics over Internet Protocol	ISO 13400	
DTC	Diagnostic Trouble Code	ISO 14229 / SAE J2012	
E2E	End-to-End Protection	AUTOSAR Standard	
ePDC		Synthetic System	
FCR	Fault Containment Region	Fault Tolerant Systems	
HIL	Hardware-in-the-Loop	Testing Standard	
HVIL	High-Voltage Interlock Loop	ISO 6469	
SOME/IP		AUTOSAR Standard	
SWC	Software Component	AUTOSAR Standard	
UCM		AUTOSAR Adaptive	
UDS	Unified Diagnostic Services	ISO 14229	

End of Specification Document

This concludes the 101-page High-Level Design (HLD) Specification for the AUTOSAR Adaptive Powertrain Domain Controller (v1.0 Comprehensive Baseline).